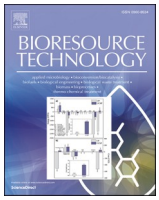
Bioresource Technology 394 (2024) 130230



Contents lists available at ScienceDirect

Bioresource Technology

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Upgrading lignin macromolecular by green and recyclable ternary deep eutectic solvents

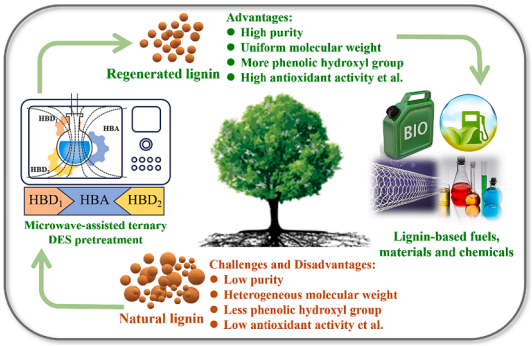
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HIGHLIGHTS

- A ternary DES was prepared, explored and compared with binary ones.
- Microwave-assisted DES pretreatment was applied for lignin processing.
- Lignin has been improved such as purity, dispersity and thermal stability.
- The interaction between ternary DES and lignin was analyzed by MD simulation.
- Dissolution mechanism of microwave-assisted ternary DES on lignin was proposed.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Microwave
Dissolution
Regeneration
Depolymerization
Molecular dynamics

ABSTRACT

Lignin is the most abundant natural aromatic macromolecule in the nature, but its high value-added utilization has been seriously hindered by the highly random and branched structures and the high difficulty in separation and purification. A microwave-assisted ternary deep eutectic solvent (DES) composed by formic acid, lactic acid and choline chloride was developed for lignin pretreatment. The effects of three types of DES on main characteristics of lignin were investigated, and the corresponding dissolution mechanism was proposed. The results showed that, the microwave-assisted ternary DES pretreatment showed an obvious improvement on main characteristics of regenerated lignin, e.g., a higher purity, lower molecular weight with reduced dispersity, improved thermal stability, higher phenolic hydroxyl content, and increased antioxidative activity in comparison with control. It is expected that the lignin macromolecular can be facile regulated and upgraded by the proposed ternary DES.

1. Introduction

Lignocellulosic biomass, as the most abundant renewable resource in the world, is mainly composed of cellulose, hemicelluloses, lignin, and a small amount of ash, as well as extractives, in which lignin accounts for about 20 %–30 % (Gu et al., 2021; Ragauskas et al., 2014). In the

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https://doi.org/10.1016/j.biortech.2023.130230
Received 25 October 2023; Received in revised form 18 December 2023; Accepted 18 December 2023
Available online 22 December 2023
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nature, lignin is the most abundant natural aromatic macromolecule (Hong et al., 2020), whose rich and abundant functional groups endow lignin with unique advantages such as UV blocking ability, antibacterial and antioxidant properties, so as to be used in the production of valuable aromatic chemicals and functional materials (Figueiredo et al., 2018; Liu et al., 2022; Morena et al., 2021; Spiridon, 2020). However, owing to the highly random and branched 3D polyphenol structures and the high difficulty in separation and purification, the high value-added and efficient utilization of lignin has been seriously hindered (Sangha et al., 2014; Zakzeski et al., 2010). In conventional pulp and paper industry, as one kind of by-product, most lignin is combusted to generate heat or electricity (Farag et al., 2014). In addition, the existence of lignin hinders the efficient utilization of carbohydrates in lignocellulosic biomass. For example, to obtain bleached pulp with high cellulose content, it is essential to remove lignin via cooking (e.g., Kraft pulping). However, most conventional methods inevitably cause severe destruction of natural lignin structure and abnormal condensation, which are not conducive to utilization of lignin (Lobato-Rodríguez et al., 2023). These processes invariably involve the use of toxic and hazardous chemicals, which affect the environment and raise process costs.

Deep eutectic solvent (DES) is composed of two or three components, i.e., hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) (Ren et al., 2023). As an emerging green solvent, DES resembles ionic liquids (ILs) in terms of physical and chemical characteristics, and has already attracted rapid and wide attentions due to its unique advantages, such as low vapor pressure, low cost of constituents, easy preparation, recyclability, and biodegradability (Cunha & Fernandes, 2018). DES can promote cleavage of labile ether bonds among phenylpropane units, which leads to depolymerization and then separation of lignin from lignocellulosic biomass, producing lignin with low molecular weight and high activity (Alvarez-Vasco et al., 2016; Shen et al., 2020).

So far, DES has been extensively studied in biorefinery, in which acidic DES using choline chloride as HBA has been widely studied due to its unique advantages such as good dissolution capacity, mild catalytic activity and high selectivity (Li et al., 2019; Tan et al., 2019). The DES composed of choline chloride and carboxylic acid would remove lignin at a relatively low temperature, and the extraction efficiency of lignin varies with changing carboxylic acid types (Chen et al. (2020); Xia et al., 2018). It is reported that the DES composed of choline chloride and lactic acid can effectively extract lignin from wood and grass (up to 93 % of lignin removal ratio); meanwhile, the obtained lignin retains unique advantages (e.g., ether bond structure and low molecular weight) (Alvarez-Vasco et al., 2016). In addition, the presence of lactic acid would increase the number of active protons and enhance the proton mobility of choline chloride-lactic acid DES, which can improve the interaction between lignocellulosic biomass and the solvents, encouraging the breakage of ester bonds in lignin and carbohydrates complexes (Yan et al., 2021). The lignin obtained from choline chloride-formic acid DES has a higher solubility and more stable thermal stability than that from choline chloride-lactic acid DES (Lynam et al., 2017). Choline chloride-formic acid DES exhibits a high swelling capacity for lignocellulosic biomass without destroying chemical structure of cellulose (Li et al., 2021c), but it has obvious damage to lignin structure (Li et al., 2021b). Although DES pretreatment on the separation and utilization of lignocellulosic biomass has made some progress, there are still some challenges need to be well addressed, such as the component design, process optimization, mechanism, and interactions between DES with lignocellulosic biomass, as well as advanced characterization. Thus, more efforts still need to devote to the synergistic effect of DES composed by different types of carboxylic acid and choline chloride.

Herein, a ternary deep eutectic solvents composed by formic acid (FA), lactic acid (LA) and choline chloride (ChCl) were synthesized at three different mole ratios, characterized, and then explored as the solvent for lignin pretreatment with the assistance of microwave. The structure and properties of the regenerated lignin (RL) were comprehensively characterized. The dissolution mechanism of microwave-assisted DES pretreatment on lignin was proposed. It is expected that the lignin macromolecular can be greatly upgraded and rationally regulated by the ternary DES pretreatment.

2. Materials and methods

2.1. Materials

Double enzymatic lignin (DEL) was chosen as the raw materials, because DEL has a similar structure with natural lignin and higher purity, which was prepared according to previous report (Wang et al., 2017). In brief, the poplar wood powder was extracted with toluene-ethanol (2:1, v/v) in Soxhlet extractor for 6 h, ball milled at 450 rpm for 5 h, and then enzymatic hydrolysis under the conditions: 4.0 % (w/w) of consistency, pH 4.8, excessive cellulose loading of 100 FPU/g substrate, 50 °C, shaking speed of 150 rpm, and duration time of 48 h. After the enzymatic hydrolysis, the mixture was centrifuged, washed with acetic acid-sodium acetate solution, and lyophilized. The above treatments of ball milling and enzymatic hydrolysis were repeated again, finally obtaining the DEL. The main chemical compositions of the as-prepared DEL were determined to be 81.02 % total lignin, 2.86 % glucose, 5.15 % xylose, and 1.56 % arabinose.

ChCl, formic acid (FA) and lactic acid (LA) are purchased from Meryer (Shanghai) Chemical Technology Co., Ltd., China. Butylhydroxyanisole (BHA), 1,1-diphenyl-2-trinitrophenylhydrazine (DPPH), dimethyl sulfoxide (DMSO), DMSO-d_6 and 1,4-dioxane were purchased from Macklin (Shanghai) Biochemical Co., Ltd., China. Nile red dye (NEt2) was purchased from Heowns Biochem Technologies Co., Ltd., China. p-nitroaniline was purchased from Kemat (Tianjin) Chemical Technology Co., Ltd., China. N, N diethyl-p-nitroaniline was purchased from Beijing Bailingwei Technology Co., Ltd.

2.2. Preparation and characterization of DES

ChCl (C) was mixed with formic acid (F) and/or lactic acid (L) at different molar ratios (based on the molar ratio of ChCl) at room temperature. Then, the mixtures were kept heating at 60 °C for 2 h with magnetic stirring until it became a uniform and stable solution. After that, the DES were obtained, and named as 1) LC (ChCl: FA: LA = 1:0:10), 2) FLC (ChCl: FA: LA = 1:1:1), 3) FLC1 (ChCl: FA: LA = 1:1:2.5), 4) FLC2 (ChCl: FA: LA = 1:1:5), and 5) FC (ChCl: FA: LA = 1:2:0). All the as-prepared DES were sealed and kept in a desiccator.

The viscosity of DES was measured at 60 °C with 1.0——1.1 mm Ubbelohde viscometer and calibrated according to the viscosity of ethanol. To further evaluate the performance of DES on pretreatment and dissolution of DEL, the solvation properties of the as-prepared DES was quantified using the Kamlet-Taft solvatochromic parameters including α, β, and π^* (Oh et al., 2020). The probes used in the experiment were Nile red dye (NEt2), p-nitroaniline, and N,N-diethyl-p-nitroaniline. The absorption spectra of dyes were determined by UV–vis spectroscopy (DR-6000, Beijing Rayleigh Analytical Instrument Co., Ltd., China). The calculation was conducted according to the following equations:

$$\nu = 1/(\lambda_{\max} \times 10^{-4}) \quad (1)$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \nu_{NR}}{1.6078} \quad (2)$$

$$\beta = 11.134 - \frac{3580}{\lambda(NH_2)_{max}} - 1.125\pi^* \quad (3)$$

$$\pi^* = 0.314(27.52 - \nu_{(NE_{t2})}) \quad (4)$$

where α is the hydrogen-bond-donating ability (i.e., Lewis acidity), β is the hydrogen-bond-accepting ability (i.e., Lewis basicity), and π^* is the

polarizability of the solvents; λ_{max} is the maximum absorption wavelength of the dye used.

2.3. DES pretreatment and regeneration of lignin

The DES pretreatment was carried out in a microwave reactor (WBFY-205, JINBO Instrument Co., Ltd., China) under the following: 30 mg DEL and 30 g DES were mixed in a round-bottom flask under microwave at 800 W for 3 min. Then, the undissolved particles were removed by 0.22 μm hydrophilic filter membrane, obtaining the lignin-DES solution. The regenerated lignin (RL) was obtained by antisolvent precipitation, that is, deionized water was added into the DEL-DES solution until the volume ratio reached 5:1, and kept continuously stirring for 2 h to precipitate the dissolved lignin. The solid part was separated with 0.22 μm hydrophilic filter membrane, washed until neutral stage was achieved, and frozen dried at –60 °C, and finally obtain the RL. Fig. 1 shows the experimental flow of DES pretreatment and regeneration of lignin.

2.4. Determination of solubility of DEL in DES

The solubility of lignin in DES was determined by stepwise dissolution method. 0.2000 ± 0.001 g of DEL was added into 10 g DES. The mixture was kept stirring at 60 °C for 1 h and dispersed under ultrasound for 30 min. The above operation was repeated until the undissolved DEL particles were observed under a microscope of 100 magnification (Sosa et al., 2020).

2.5. Characterization of regenerated lignin

The composition of DEL and RL was analyzed according to the standards (NREL/TP-510–42618) to reveal the purity of lignin. FTIR (650, Tianjin Gang dong Sci. &Tech. Development Co, Ltd., China) was employed to analyze the chemical features of RL (Wang et al., 2017). The structure of RL was further characterized by 2D-HSQC NMR spectrometer (Bruker AVIII 400 MHz, Swiss Brooke Company, Switzerland) at 25 °C in DMSO-d_6. The thermal stability of lignin was determined by synchronous thermal analyzer (SDT650, American Tower, USA). The morphology of lignin was observed and analyzed by scanning electron microscopy (JSM-IT300LV, Japan Electronics Co., Ltd., Japan). The particle size of RL was determined by laser particle size analyzer (90Plus, Brookhaven Instruments, USA). The molecular weight values (weight average, M_w) and number average (M_n) of the RL samples were determined by gel permeation chromatograph (Model 3326985, Fairborn Precision (Shanghai) Instrument Co., Ltd., China) (Wang et al., 2020). The UV–vis of RL was measured by Ultraviolet visible spectrophotometer (UV–vis, Beijing Rayleigh Analytical Instrument Co. Ltd., China).

The phenolic hydroxyl groups content (PhOH) in lignin was determined by ultraviolet differential spectroscopy according to the literature (Tan et al., 2019). In brief, 1.0 mg of DEL was dissolved in a mixture of 5.0 mL 1,4-dioxane and 5.0 mL 0.2 M NaOH solution. The solution was filtered through a 0.45 μm pinhole filter to remove the insoluble solids. Two parts of 2.0 mL filtrate were taken, one of which was diluted to 0.08 g/L with a pH = 6.0 buffer solution as the reference solution, the other part was diluted with 0.2 M NaOH solution in the same proportion, and the absorbance values at 300 nm and 360 nm were measured, respectively. The PhOH was calculated according to the following formula (5)–(6):

$$\Delta a = \frac{D}{c \cdot l} \tag{5}$$

$$PhOH = \left(0.250 \times \Delta a_{300,NaOH}\right) + \left(0.107 \times \Delta a_{360,NaOH}\right) \tag{6}$$

where D and c represents the absorbance and concentration of lignin solution, respectively; and l is the thickness of cuvette.

The antioxidant property of RL was measured in accordance with a previous report (Guo et al., 2020). Briefly, a certain amount of lignin samples (0.05 to 2.5 mg) were dissolved in a water/dioxane (1:9, v/v) solutions. Then, 3.9 mL of DPPH-ethanol solution (25 mg/L) and 0.1 mL of the lignin sample solution were thoroughly mixed and kept shaking at 25 °C for 0.5 h. The UV–vis spectrophotometer was used to measure the absorbance at 516 nm of the above solution. The DPPH free radical clearance (*IP*) was obtained using the following formula. The antioxidant activity of lignin was indexed and denoted as IC50 (i.e., the lignin concentration at which the inhibition level reaches 50 %), as follows:

$$IP = \frac{A1 - A2 + A0}{A1} \times 100\% \tag{7}$$

where A_0, A_1, and A_2 represent the absorbance at 516 nm of the blank sample, DPPH solution, and the mixed solution after reaction, respectively.

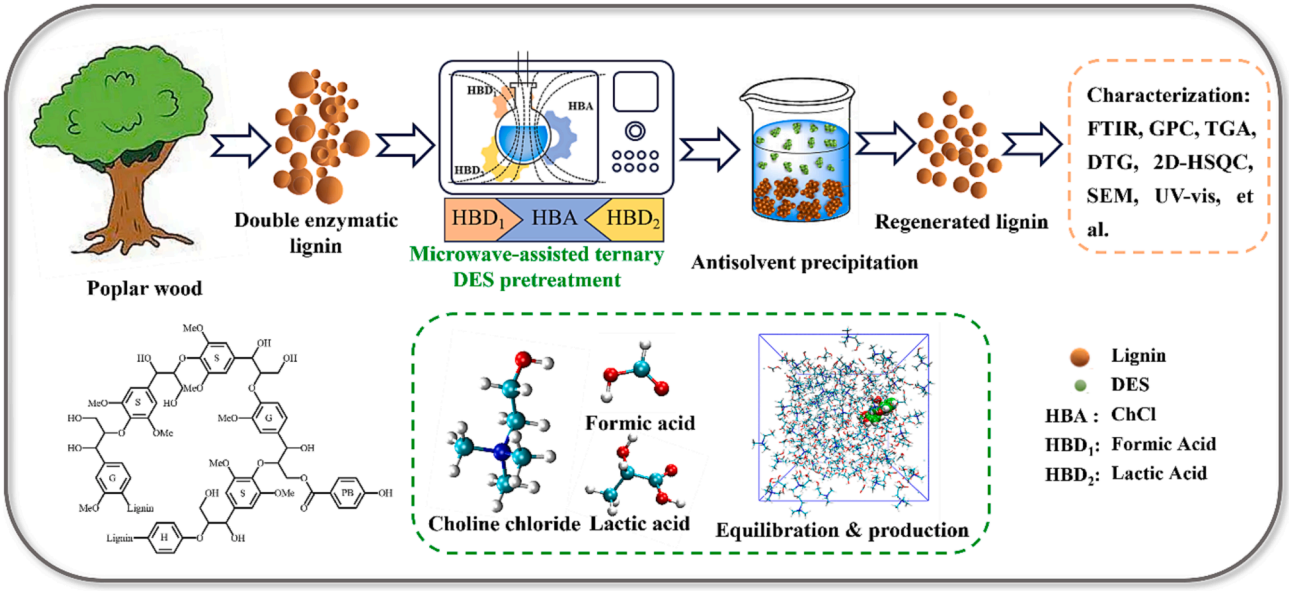


Fig. 1. Experimental flow of microwave-assisted DES pretreatment and regeneration of lignin.

All these above determination was performed in triplicate, and the results were analyzed and reported at a confidence level of 95 %.

2.6. Computational analysis

To further reveal the dissolution mechanism of DES pretreatment on lignin, and provide information for the selection and rational design of DES, molecular dynamics (MD) analysis were performed and analyzed. Veratrylglycerol-b-guaiacyl ether (VG) was chosen as the model lignin. Three MD system were built, that is, 1) VG-LC system contained 1 VG molecule, 23 ChCl molecules, and 232 LA molecules; 2) VG-FLC system contained 1 VG molecule, 85 ChCl molecules, 85 FA molecules, and 85 LA molecules; 3) VG-FC system contained 1 VG molecule, 85 ChCl molecules, and 170 LA molecules. MD simulation was performed through Gromacs2022.1 software (Van Der Spoel et al., 2005). The equilibrium simulation was performed in the NPT ensemble at 298 K and 1 bar for 1 ns. The production simulation was carried out in the NPT ensemble at 298 K for 50 ns. The hydrogen bond number and radial distribution functions (RDF) were calculated using Gromacs2022.1. Quantum chemistry calculation was performed with a Gaussian16 software to investigate the interaction between ions or molecules. VMD software was used to visualize the systems and obtain the conformation of the complex molecules.

3. Results and discussion

3.1. Characterization of DES

The main characteristics of binary DES and ternary DES composed of formic acid (F), lactic acid (L) and choline chloride (C) are shown in Table 1. Firstly, Kamlet-Taft parameters have been used to evaluate the performance of DES on the pretreatment and dissolution of lignocellulosic biomass, because the performance of DES is mostly depend on its solvation ability. In this section, the solvation properties of the as-prepared DES was evaluated by Kamlet-Taft solvatochromic parameters, i.e., the hydrogen bond giving capacity (α), hydrogen bond accepting capacity (β) and polarizability (π^*). All these types of DES exhibit similar β values because they all use choline chloride as HBA. Among all these types of 'DES, the hydrogen bond giving capacity (α) of FC was the highest (i.e., 1.29), while that of LC was the lowest (i.e., 0.98). The polarizability of LC was the highest (π^*=1.165), while that of FC was the lowest (π^*=1.111). As for the ternary DES composed of formic acid, lactic acid and choline chloride, α values were reduced but π^* were increased with introducing lactic acid. It was found that the DES for dissolution of Kraft lignin exhibited a positive linear relationship with α and π^* instead of β (Zhang et al., 2023). It was also reported that the DES with a higher hydrogen bond giving capacity (α) is conductive to catalyzing the cleavage of ester and ether bonds in lignin during pretreatment (Provost et al., 2022). Thus, FC and FLC would be considered as good solvents for the pretreatment. However, the acidic DES may lead to an excessive degradation and condensation of lignin (Li et al., 2021a). Thus, it is necessary to reasonably optimize the components of DES.

Secondly, the viscosity values at 60 °C of the above five types of DES were determined. As is all known, viscosity is one of the most important factors affecting DES pretreatment and dissolution of lignocellulosic biomass. Thus, it has already become one major obstacle for the potential industrial applications of DES. An appropriate viscosity is conductive to improving pretreatment efficiency. The viscosity values of these DES samples were in the range of 34.7–41.2 mPa•s, in which the viscosity of FC was the lowest while that of LC was the highest. In comparison with the viscosity of LC, the viscosity values of the ternary DES composed of FA, LA and ChCl were gradually reduced with increasing FA content. It can be seen that the viscosity of the ternary DES composed by formic acid, lactic acid and choline chloride is tunable, which can be controlled by changing their mole ratios.

Finally, the lignin solubility in these types of DES is determined. The results showed that, the lignin solubility in FC was the highest (26.37 g/100 g DES), the lignin solubility in LC was the lowest (20.83 g/100 g DES), and that in ternary DES were in between the two binary DES (22.45——25.72 g/100 g DES), which were extremely higher than the required one in the pretreatment process (3.00 g/100 g DES). This result is consistent with the obtained Kamlet-Taft parameters. It was also reported that the lignin solubility was linearly correlated to the $\alpha - \beta$ values of ChCl-acid DES (Zhang et al., 2023). Thus, all these types of DES (FC, LC and FLC) would be used for the pretreatment and dissolution of lignin, in which FLC is further explored since it exhibits advantages of both FC and LC.

Table 1
The main characteristics of DES samples.

DES		Ratio (FA: LA: ChCl)	Kamlet-Taft parameters π^*	α	β	Viscosity at 60 °C (mPa•s)	Solubility of lignin (g/100 g)
LC		0:10:1	1.165	0.98	0.53	41.2	20.83
FLC	FLC	1:1:1	1.147	1.23	0.56	36.8	25.72
	FLC1	1:2.5:1	1.129	1.13	0.57	38.1	24.63
	FLC2	1:5:1	1.142	1.17	0.54	39.5	22.45
FC		2:0:1	1.111	1.29	0.54	34.7	26.37

3.2. Characterization of regenerated lignin

3.2.1. Yield and purity of regenerated lignin

Fig. 2a, b showed the yield and purity of regenerated lignin (RL) obtained from different types of DES. The yield of RL with three types of DES was higher than 77 %, which was mainly because the as-prepared DES can effectively break the hydrogen bonds and other chemical bonds in lignin. The ability of FLC as a ternary DES to pretreat lignocellulosic biomass was also demonstrated. It can be seen that the RL yield of FC pretreatment was lower than that of LC and FLC, which may be due to the high acidity of FC, leading to the depolymerization of lignin into small molecules.

The purity of RL samples including FC-L, FLC-L and LC-L were determined to be 95.5 %, 94.9 %, and 91.3 %, increased by 18 %, 17 %, and 13 % in comparison with that of DEL (81.0 %), respectively, indicating that DES pretreatment greatly improved the purity of RL. The purity of FC-L was much higher than that of LC-L because of the good solubility of lignin in FC, which was consistent with the solubility and Kamlet-Taft parameters. Thus, it can be seen that the lignin samples obtained from three types of DES pretreatment have high yield (>77 %) and a high purity (>91 %), indicating a good possibility of these DES pretreatment for lignin. It was worth pointing out that the purity of FLC-L was very close to that of FC-L, indicating that FLC, a ternary DES, has a great potential in lignin pretreatment.

3.2.2. FTIR analysis of regenerated lignin

FTIR is used to study the structural changes of regenerated lignin obtained from different types of DES. As can be seen from Fig. 2c, d that, the characteristic peaks of the three regenerated lignin samples were consistent with that of DEL, which indicates that the main structure of lignin macromolecules did not change significantly. The characteristic peaks can be seen from these RL samples. The absorption peaks at 1605 cm^{-1}, 1508 cm^{-1} and 1459 cm^{-1} can be assigned to the stretching vibration of lignin aromatic benzene ring skeleton. The peak at 1421 cm^{-1} can be attributed to the stretching vibration of methyl and methylene (Li et al., 2021b). Both the peaks at 1122 cm^{-1} and 1327 cm^{-1} can be attributed to the aromatic C–H bond and C–C bond in S unit, respectively, which are the characteristic signals of hardwood lignin. As for the RL, the peak intensity of aromatic C–H bond in RL was increased, indicating that S-type lignin structure was degraded to a certain extent. The peak at 2940 cm^{-1} can be assigned to the stretching vibration of C–H in methyl and methylene groups, and the peak intensity in FC-L was

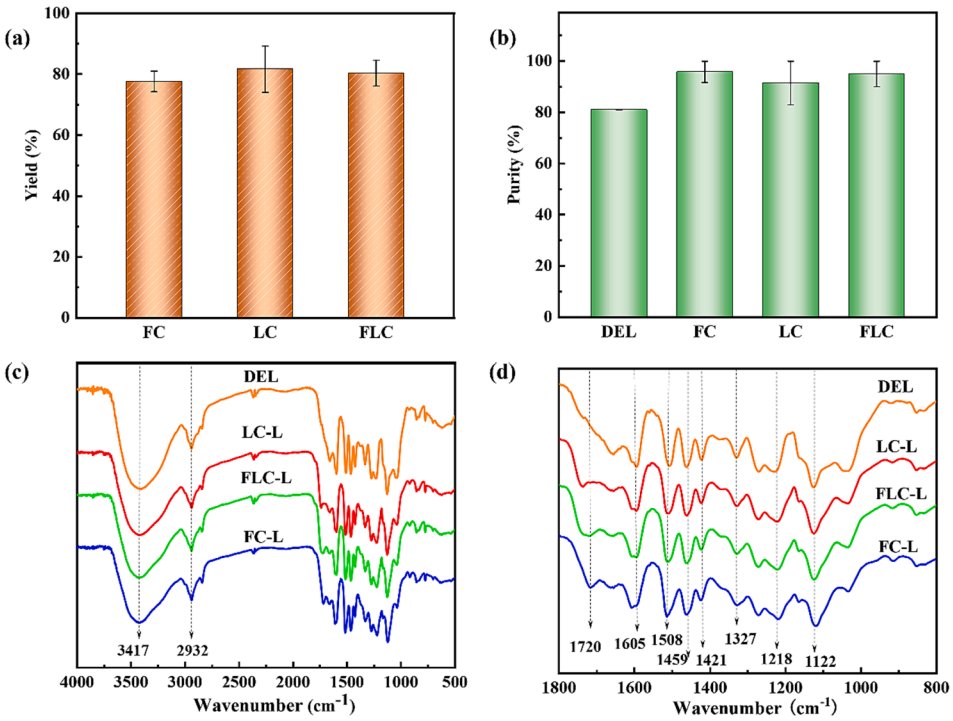


Fig. 2. Yield (a), purity (b) and FTIR (c, d) of regenerated lignin.

slightly reduced. The peak around 1720 cm^{-1} can be attributed to the stretching vibration of C = O in the conjugated carbonyl group, the peak intensities of these RL samples were significantly improved, indicating the esterification of lignin occurred during the lignin dissolution by DES.

3.2.3. 2D-HSQC NMR analysis of regenerated lignin

2D-HSQC NMR were conducted to reveal the structural changes of lignin samples treated with DES. The aliphatic–oxygenated and aromatic regions of lignin samples was shown in Fig. 3a, respectively. The 2D-HSQC spectra of aliphatic region of lignin samples can provide the characteristic signals of the connection between basic structural units of lignin (Guo et al., 2022). The quantity of S and G units in lignin samples can be indicated by the strong signal of methoxy groups at δ_C/δ_H 56.2/3.73 ppm. The correlation signals of *β–O–*4 ethers (A), resinols (B), and phenylcoumarins (C) can be observed. The signals at δ_C/δ_H 72.47/4.93 can be attributed to C–H (α position) of *β–O–*4 ethers (A_α). The obvious signals at δ_C/δ_H 84.11/4.40 and 85.20/4.10 ppm can be corresponded to the C–H (*β* position) of *β–O–*4 ethers linked G and S unit ($A_{\beta(G)}$ and $A_{\beta(S)}$). The signal at δ_C/δ_H 59.9/3.50–3.63 ppm belongs to the C–H of *γ* position in the side chain of *β–O–*4 ethers (A_γ). The signal at δ_C/δ_H 63.29/4.33–4.45 ppm belongs to the C–H of *γ* position in the side chain of the *γ*-acetylated *β–O–*4 linkages (A'_γ). The presence of resinols (*β–β*) was further confirmed by three different C–H related signals (*α*, *β*, and *γ* positions) at δ_C/δ_H 84.8/4.60, 53.5/3.06, and 71.87/3.82–4.25 ppm, respectively. It is worth pointing out the discovery of cross-signals of phenylcoumarin (*β–*5) in the aliphatic-oxygenated regions of lignin. The C–H correlation signal of *γ* position in the phenylcoumarans appears at δ_C/δ_H 62.71/3.45 ppm (Shen et al., 2016; Sun et al., 2021). More significantly, after DES pretreatment, the signals for *β–O–*4 bond of lignin were diminished or vanished, and the signals entirely disappeared in FC-L. The amounts of *β–O–*4 ethers in DEL, LC-L, FLC-L, and FC-L were determined to be 60.42, 23.32, 18.50, and 3.54/100 Ar, respectively. This indicated that the *β–O–*4 ether of DEL was greatly destroyed during the DES pretreatment. Among them, FC generated the strongest damage to DEL structure. It was obvious that the signals of $C_\beta - H_\beta$ (B_β) in the three types of RL samples were much stronger than that of DEL, further confirming that the changes of lignin during DES pretreatment. In addition, the RL exhibited a distinct acylation of *β–O–*4 bond (A′), which was consistent with FTIR analysis (Fig. 2c,d). This was mostly the consequence of the lignin being acylated by acid DES (Shen et al., 2019).

As shown in Fig. 3b, in the aromatic region, the signals of DEL and RL can be clearly distinguished in the spectra. The main signal of the C − H bond in S units (2, 6 positions) appeared at δ_C/δ_H 103.4/6.71 ppm. The salient signal for the C–H correlation in C_α-oxidized S unit (S′) (2, 6 positions) appeared at δ_C/δ_H 106.91/6.82 ppm. The C–H correlated signals in aromatic rings of G unit (2, 5, and 6 positions) were found at δ_C/δ_H 111.16/6.99, 116.25/6.78, and 119.62/6.80 ppm, respectively. Therefore, it can be concluded that DEL and RL samples were predominantly comprised of S and G units, which were identical to the ordinary hardwood. Additionally, p-hydroxybenzoates units (PB) was identified in the spectra, in which the related signal for the C–H (2, 6 positions) appeared at δ_C/δ_H 130.4/6.71 ppm. After DES pretreatment, the chemical shift of G units occurred because of the alteration of electron distribution of the aromatic ring in G units resulted from the acid-catalyzed condensation to create new carbon–carbon bonds (Torr et al., 2012). In comparison with DEL, the signals' intensity of the condensed structures in RL samples were obviously increased, indicating the occurrence of the condensation during DES pretreatment. The signal corresponded to $S'_{2,\ 6}$ structural unit was present in FC-L, but nearly vanished in LC-L and FLC-L, indicating that the condensation of lignin at the $S'_{2,\ 6}$ position occurred as a result of the pretreatment with DES containing lactic acid. The results showed that the condensation of G type structural units in *β–β* position occurred. It was also found that all RL samples showed different levels of condensation due to the appearance of condensed $S_{2,\ 6}$ and G_2. The S/G ratio is also a significant indication for the structural changes during DES pretreatment (Wang et al., 2020). After DES treatment, the S/G ratio of RL samples were increased, because G units were partially depolymerized and then removed with preference (Torr et al., 2012).

In addition, the phenolic hydroxyl groups' content in lignin was determined. As can be seen from Table 2, the content of phenolic hydroxyl groups in LC-L, FLC-L, and FC-L was measured to be 1.87, 2.14, and 2.49 mmol/g, respectively, which were much higher than that in DEL (1.18 mmol/g). After pretreatment with DES containing different

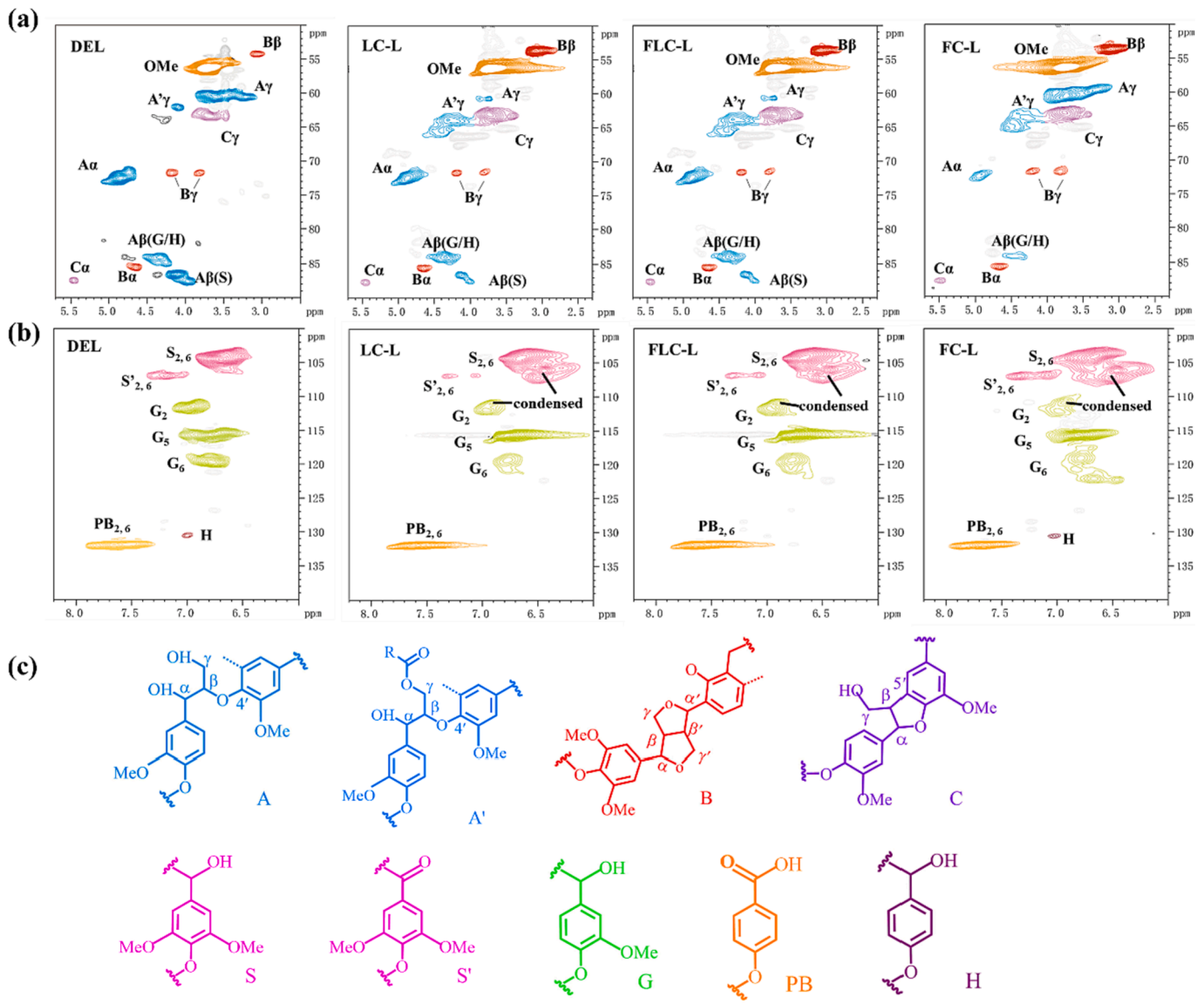


Fig. 3. Aliphatic (a) and aromatic region (b) 2D-HSQC NMR spectra of the RL and main structures in lignin samples (c) (A, *β–O–*4 linkages; A′, γ-acetylated *β–O–*4 linkages; B, resinol formed by *β–β* linkages; C, phenylcoumarans formed by *β*–5 and α–*O*–4 linkages; S, syringyl units; S′, oxidizing syringyl units; G, guaiacyl units; PB, parahydroxybenzoic acid; H, *para*-hydroxy-phenyl units).

Table 2
Quantitative analysis of main structure units in lignin samples.

Lignin samples	S/G	*β–O–*4 (%)	Phenolic hydroxyl groups (mmol/g)
DEL	1.74	60.42	1.18
LC-L	3.16	23.32	1.87
FLC-L	7.25	18.50	2.14
FC-L	10.57	3.54	2.49

formic acid content, the higher the formic acid content, the higher the phenolic hydroxyl groups' content in RL, which was mainly because the increased damage of *β–O–*4 linkage in lignin. The increase of phenolic hydroxyl groups' content would endow lignin with a much higher antioxidative activity, which is conducive to further high value-added utilization of lignin.

3.2.4. *TGA analysis of regenerated lignin*

As shown in Fig. 4a, b, the thermal degradation process of lignin samples can be divided into three stages: (1) Below 200 °C, all lignin samples had rapid weight loss at 100 °C, which was mainly due to the removal of water molecules that form hydrogen bonds with the hydrophilic groups in lignin; the peak at 130–200 °C was attributed to the dehydration of lignin hydroxyl groups and the breaking of weak C–O bonds in *β–O–*4 structure; (2) The main degradation occurred in the range of 200–500 °C, which was mainly due to the pyrolysis of the connecting bonds of structural units such as aryl ether bonds in lignin, and the rapid degradation was mainly attributed to the oxidation of lignin. It can be seen that DEL, LC-L, FLC-L and FC-L showed obvious shoulder peaks at 298 °C, 256 °C, 259 °C and 257 °C, respectively, this difference was mainly due to the difference of carbohydrates contents in lignin samples; Within the range of 330–367 °C, these four lignin samples reached the highest degradation rate, i.e., DEL (331 °C), LC-L (360 °C), FLC-L (364 °C), and FC-L (367 °C), respectively, which was consistent with the purity of lignin samples, indicating the RL had a much higher thermal stability; (3) Above 500 °C, the strong C–C bonds and some aromatic ring structures in lignin were slowly degraded; Finally, after 800 °C, the residual carbon content was in the following order: FC-L > FLC-L > LC-L > DEL, among which FC-L ranked the first followed by FLC-L. In conclusion, the DES pretreatment significantly improved the thermal stability of lignin samples mainly via increasing

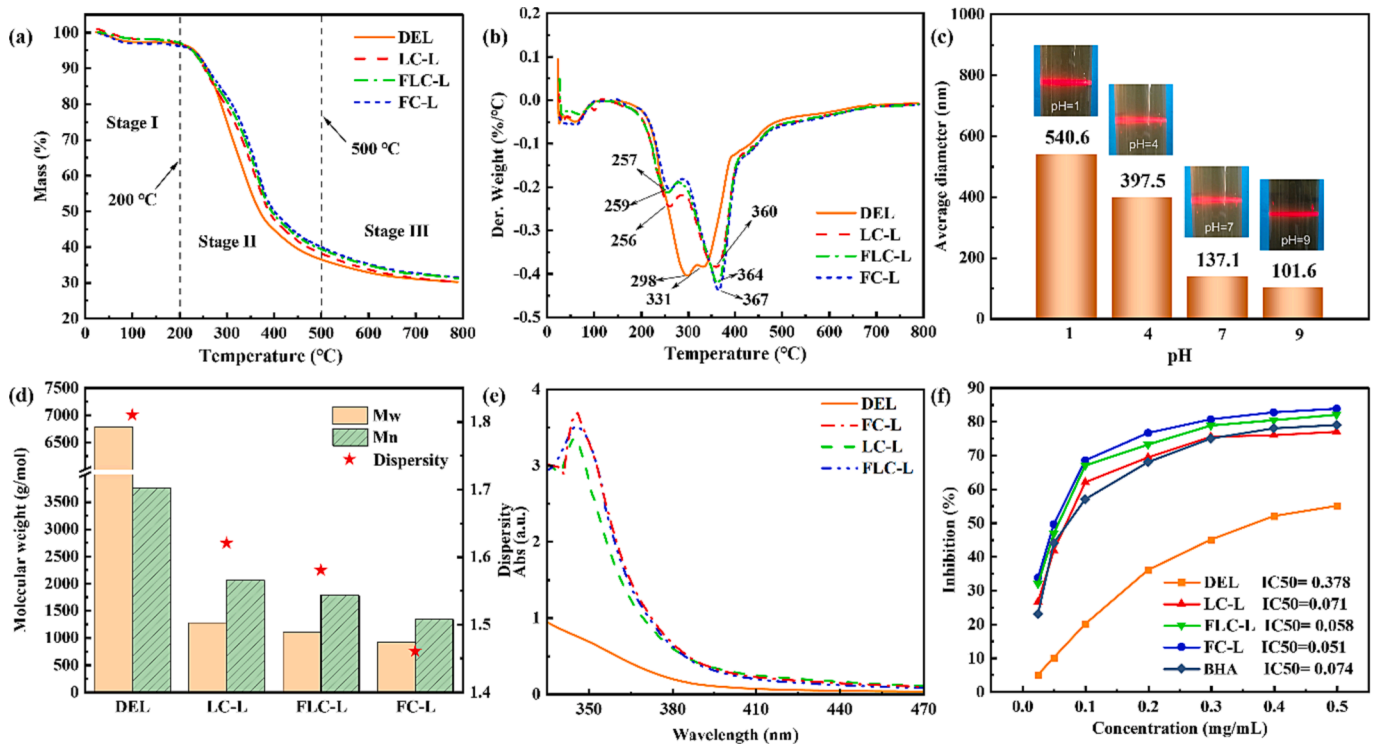


Fig. 4. TGA (a), DTG (b), molecular weight and dispersity (c), average particle size and Tyndall effect at different precipitation pH (d), UV–vis absorption spectra (e), and antioxidant activity (f) of lignin samples.

lignin purity.

3.2.5. Particle size and distribution of regenerated lignin

The effects of ternary DES (i.e., FLC) pretreatment on particle size and morphology of RL under different heating methods with conventional water bath and microwave were compared. The RL obtained from water bath heating had an average particle size of 251.6 nm and a wide range of particle size distribution, while the RL under microwave heating had an average particle size of 137.1 nm and a relatively narrow particle size distribution, which was consistent with the SEM observation (see supplementary materials). It was found that the RL obtained by acidic DES treatment was more sensitive to the pH during the anti-solvent precipitation process. Thus, FLC-L was chosen to see the effects of pH in anti-solvent precipitation process on RL. As can be seen from Fig. 4c, the particle size of the RL gradually decreased at a higher pH value during the anti-solvent precipitation. At a lower pH, the particle size was larger due to agglomeration of lignin. The ionization of carboxyl groups was dominant with increasing pH, which enhanced the repulsive force between lignin particles, reducing the particle size of RL (Chen et al., 2018)., The deprotonation occurs at pH = 9.0, the synergistic effect of the ionization of phenolic hydroxyl groups and carboxyl groups will cause lignin decomposition (Muley et al., 2019). When pH > 9.0, the dissolution of lignin further reduced the particle size. The lignin nanoparticles can be stably kept for a certain duration time, due to the strong van der Waals force and π-π superposition of hydrophobic groups. Thus, when exploring DES regenerated lignin as a nanomaterial, dissolution conditions such as heating method, temperature, and pH of anti-solvent precipitation cannot be ignored.

3.2.6. Molecular weight and UV-blocking ability of regenerated lignin

Fig. 4d showed the molecular weight and dispersity of RL samples obtained from different DES pretreatment. It can be seen that, the number-average molecular weight (M_n), weight-average molecular weight (M_w), and dispersity of RL were reduced in comparison with DEL, which indicated the lignin depolymerization in DES dissolution. FC-L had the lowest M_n and M_w, which was not conductive for lignin application, indicating the formic acid-type DES had a serious damage to lignin. On the contrary, LC-L had a much higher molecular weight, indicating the damage of LC to lignin structure was a little bit weaker than FC. It should be point that, the effect of FLC on molecular weight of lignin stranded between FC and LC. The similar trend of dispersity was also found, but lower dispersity was good for the processing of lignin. Fig. 4e showed the UV–vis spectra of RL samples in dioxane. The results showed that, compared with DEL, the absorbance of RL samples increased in both visible and ultraviolet regions, in particular the absorbance in UV region was significantly improved, which was mainly attributed to the increased phenolic hydroxyl groups' content in RL samples, indicating an enhanced UV-blocking ability of RL (Barsberg et al., 2003).

3.2.7. Antioxidant activity of regenerated lignin

The antioxidant activity indexed by DPPH of lignin samples were studied in comparison with one commercial antioxidant (BHA) (Fig. 4f). The results showed that, the free radical scavenging ability of all lignin samples gradually increased when the lignin concentration increased from 0.01 to 0.50 mg/mL; The antioxidant activity IC50 value of DEL was determined to be 0.378 but that of these RL samples were much lower than DEL, and the antioxidant activities of these RL samples were in the order of FC-L > FLC-L > LC-L, which was mainly attributed to a higher phenolic hydroxyl groups' content in FC-L compared with others. This was consistent with the phenolic hydroxyl groups' content in lignin in the previous section. As we know, the content of phenolic hydroxyl group in lignin is positively correlated with the antioxidant activity. Thus, it is suggested that the RL samples had a much better antioxidant activity than DEL. It can be seen the antioxidant activities of RL samples were much higher than or equal to that of commercial antioxidant BHA, indicating that the RL obtained from acidic DES dissolution would be a suitable raw material for commercial antioxidants.

3.3. Interaction of VG with DES systems

Fig. 5a showed the MD computational workflow. The number of hydrogen bonds has always been a crucial indicator to describe the interaction between model lignin and DES system. Fig. 5b showed the hydrogen bonding creation capability and binding energies between the three types of DES systems and the VG. The results showed that the number of hydrogen bonds of the three types of DES and VG were determined to be VG-FC (14.06), VG-FLC (12.89), and VG-LC (11.77), respectively, which indicated that the ternary DES (FLC) fell in between the other two binary DES (FC and LC). The binding energy between the three types of DES and VG was further examined. The interaction energy between the three types of DES and VG is mainly van der Waals energy (E_{vdW}). The results showed that the absolute value of E_{vdW} between FC and VG was the highest (254.21 kJ/mol), and that of the FLC and VG was in between FC and LC, which was compatible with the findings through this study. Fig. 5c showed the radial distributions of VG with the three types of DES, i.e., *g(r)*. The RDF plots of the three types of DES with VG showed that, the *g(r)* was in the order of VG-FC > VG-FLC > VG-LC, indicating that a stronger interaction between FC and VG was formed. The MD analysis was consistent with the above analysis. Thus, it can be concluded that the interaction intensity between ternary DES and model lignin can be realized by varying the proportion the components of DES.

3.4. Proposed mechanism of lignin dissolution in microwave-assisted ternary DES

Based on above analysis, the chemical transformation of lignin in microwave-assisted ternary DES dissolution process was proposed (Fig. 6). During the microwave-assisted ternary DES pretreatment, the basic structure of lignin did not change; G-type lignin can be selectively and preferentially degraded; The *β–O–*4 ether bond of lignin was broken and formed phenolic hydroxyl groups, and the damage severity of the *β–O–*4 bond increased with introducing formic acid; The aryl ether bond and C–C bond of lignin were broken, reducing the molecular weight; The hydroxyl group on the side chain of the lignin underwent a dehydration reaction, thus a portion of the lignin will go through a condensation; In addition, the hydroxyl groups on the lignin's *β–O–*4 bond will be acylated during the ternary DES processing.

4. Conclusion

A microwave-assisted ternary DES composed by formic acid, lactic acid and choline chloride strategy was developed for lignin pretreatment. This strategy exhibited an obviously positive effects on the improvement of lignin properties, such as yield, purity (>94 %), molecular weight and dispersity, thermal stability (T_{max} = 364 °C), functional groups (e.g., phenolic hydroxyl groups was increased by 81 %), and antioxidant activity. In addition, by adjusting the components and their ratio of DES, the lignin structure and properties can be rationally regulated. Finally, the dissolution mechanism of the microwave-assisted ternary DES on lignin dissolution was proposed. This strategy would pave ways for lignin utilization.

CRediT authorship contribution statement

Wei Liu: Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Zhan Li:** Writing – original draft, Visualization, Validation, Investigation, Data curation. **Qian Ren:** Visualization, Validation, Resources, Investigation, Data curation. **Chuang Jiang:** Visualization. **Jinlong Feng:** Visualization. **Qingxi Hou:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

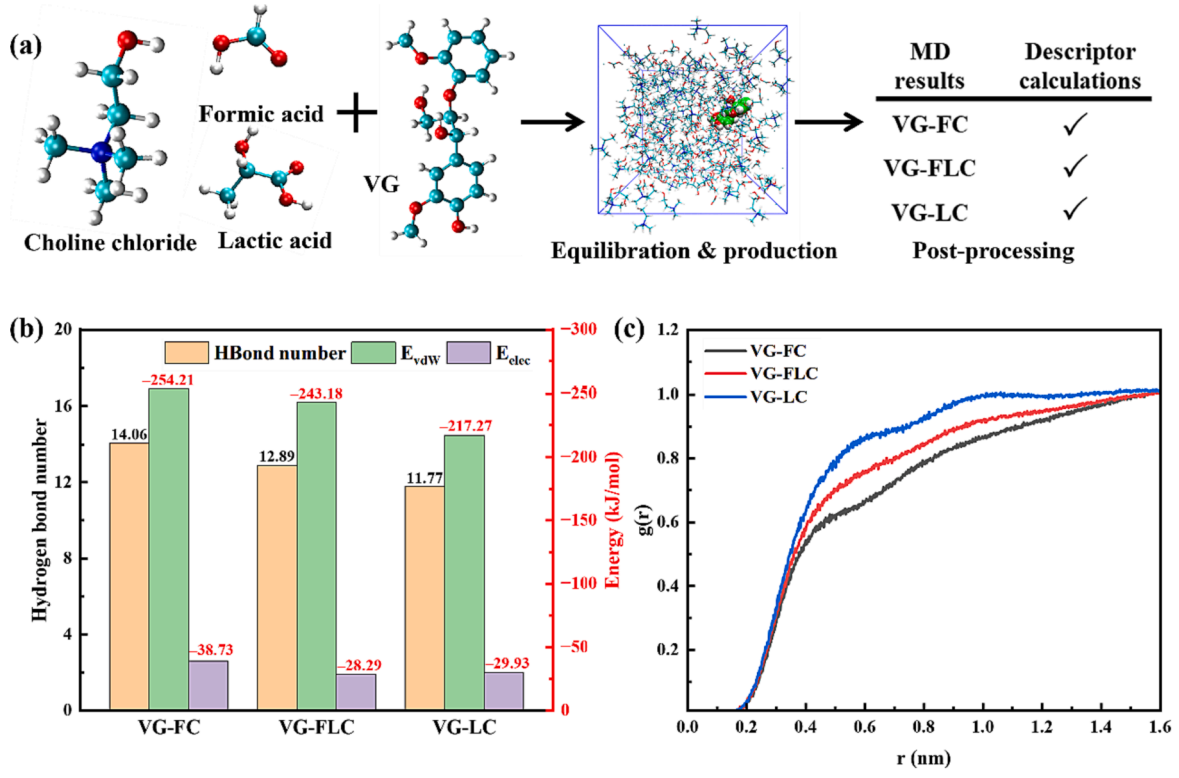


Fig. 5. MD molecular simulation (a), number of hydrogen bonds and interaction (b), and RDF (c) of VG-DES systems.

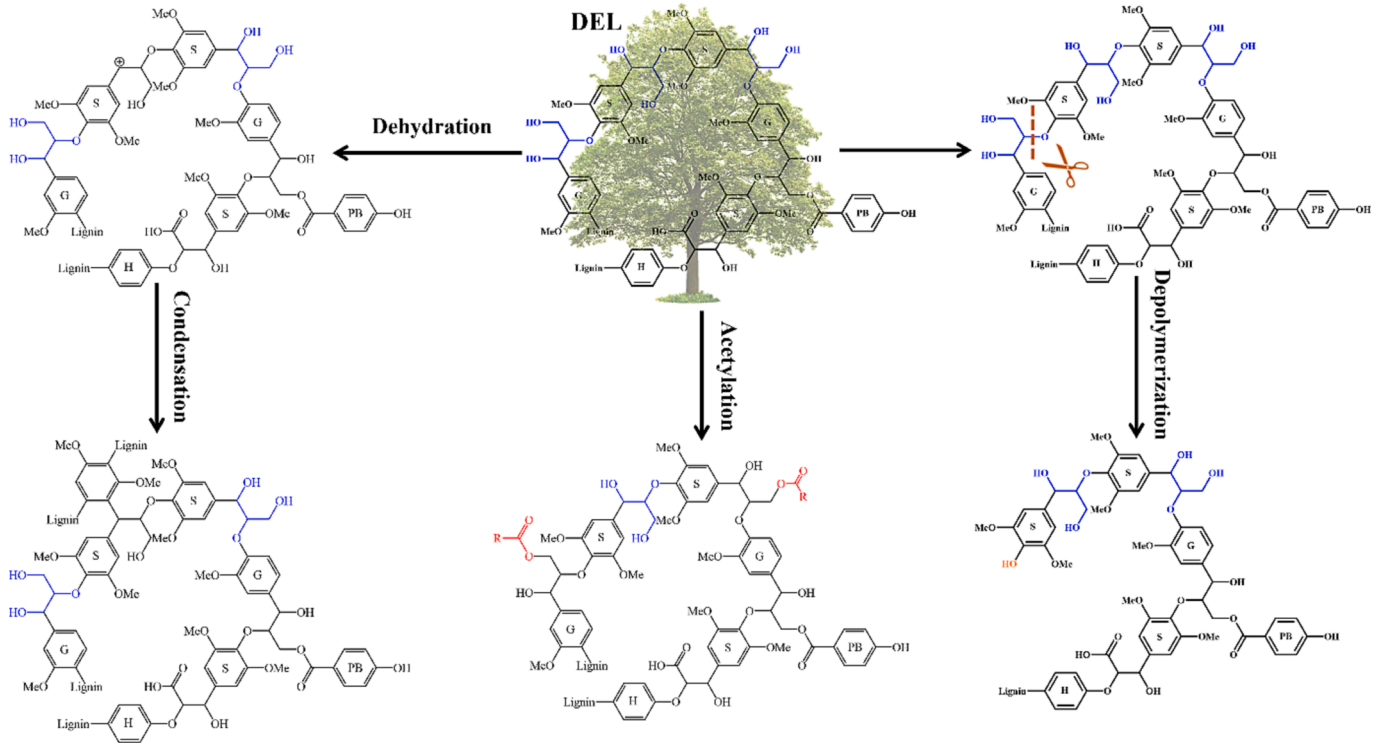


Fig. 6. Proposed dissolution mechanism of DEL in ternary DES.

Data availability

Data will be made available on request.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (32071715), Natural Science Foundation of Tianjin, China (22JCZDJC00560), and National Science Foundation for Post-doctoral Scientists of China (2019M651050).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.130230.

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